AI Agent Journey for an Experienced Java Developer

Target Outcome

Build AI agents for code generation, testing, DevOps automation, incident response, documentation, release management, and enterprise knowledge assistants.

Phase 1: AI Fundamentals (2-3 Weeks)

LLMs, transformers, tokens, embeddings, prompt engineering, RAG, fine-tuning vs prompting, context windows, function calling. Resources: DeepLearning.AI, Andrew Ng courses, Microsoft Learn Azure OpenAI. Build FAQ Bot and Documentation Assistant.

Phase 2: Python for AI (4 Weeks)

Python basics, FastAPI, asyncio, Pandas, NumPy, Jupyter, virtual environments. Build REST APIs and document analyzers.

Phase 3: Vector Databases and RAG (4 Weeks)

Embeddings, semantic search, chunking, Pinecone, ChromaDB, Azure AI Search, hybrid search. Build a DevOps Knowledge Assistant.

Phase 4: Agent Frameworks (6 Weeks)

LangGraph, Semantic Kernel, CrewAI, AutoGen, MCP. Build a DevOps Agent that reads tickets, generates plans, code, tests, and PR descriptions.

Phase 5: LLM Application Development (6 Weeks)

OpenAI APIs, Azure OpenAI, function calling, structured outputs, tool usage, memory, workflows. Build an Engineering Copilot.

Phase 6: Azure AI Ecosystem (4 Weeks)

Azure OpenAI, AI Foundry, AI Search, Azure Functions, AKS, Container Apps, monitoring, Prompt Flow. Study AI-102 topics.

Phase 7: AI Agents for Software Engineering (8 Weeks)

Build Test Automation Agent, Code Review Agent, Release Agent, and Incident Management Agent.

Phase 8: Advanced Agentic AI

Reflection agents, planning agents, multi-agent systems, human-in-the-loop, MCP, AI governance, AI security, A2A communication.

Recommended Stack

Java Spring Boot, FastAPI, Azure OpenAI, Semantic Kernel, LangGraph, Azure AI Search, Docker, AKS, Azure Monitor, Grafana, Prometheus.

Certifications

AI-102 Azure AI Engineer, Azure OpenAI learning paths, Microsoft Applied Skills for Azure OpenAI and Azure AI Search.

12-Month Roadmap

Months 1-2: Fundamentals and Python. Months 3-4: RAG and Search. Months 5-6: LangGraph and Semantic Kernel. Months 7-8: Azure OpenAI and MCP. Months 9-10: Agents and Multi-Agent Systems. Months 11-12: Production-grade DevOps and Test Automation Agents.